

SAMIDHA DHAWALE

AI ML Engineer and Full Stack Developer

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Profile Summary

Computer Engineering student (AIML Honors) focused on **LLM/RAG systems, NLP, and applied ML**. Built RAG and Transformer-based systems from scratch, with experience across **model development, backend APIs, and deployment**.

Technical Skills

Languages: Python, JavaScript, SQL, Java, C++

Frontend: HTML, CSS, React.js, Tailwind CSS, Bootstrap, React Native

Backend: FastAPI, Node.js, REST APIs

Databases: MySQL, PostgreSQL, MongoDB

AI/ML: PyTorch, Transformers, FAISS, OpenCV, NumPy, Pandas, Matplotlib, Hugging Face

Deployment: Vercel, Hugging Face Spaces, Render, Docker, Streamlit, AWS

Tools: VS Code, Git, GitHub, Figma

Projects

Scholar Press – AI Research Paper Analyser | *React, FastAPI, FAISS, MCP* | GitHub 2026

- Built a **RAG pipeline** using **MiniLM embeddings** and **FAISS** for session-isolated document indexing.
- Refactored the comparison pipeline into a **planner worker multi-agent architecture**, coordinating **groups of 5** papers across parallel worker agents and merging results in a single synthesis pass scaling analysis to **50+ papers** in **11 LLM calls**.
- Exposed retrieval, comparison, methodology extraction, and OKF evaluation as **MCP (Model Context Protocol) tools** using **FastMCP**, enabling agent-callable access via any MCP-compatible client.
- Built an **OKF evaluation layer** using 7 domain files and **ThreadPoolExecutor**, achieving **70% faster** processing.

Mini-GPT – Transformer Built From Scratch | *PyTorch, Python* | GitHub 2026

- Built a **decoder-only GPT** from scratch in PyTorch, implementing **self-attention, multi-head attention, causal masking, positional encoding**, and **Add&Norm** blocks without `nn.Transformer`.
- Trained a **2.69M-parameter** character-level model on **1M+ Shakespeare tokens** using a custom tokenizer and batcher, with 6 layers, 6 heads, and $d_{model} = 192$, reducing validation loss from **3.71 to 1.56** in 3,000 steps.
- Implemented **autoregressive text generation** with checkpoint loading and temperature-based sampling.

MoodifyAI – Music Recommendation Engine (NLP) | *React, FastAPI, Scikit-Learn* | GitHub | Live 2025

- Fine-tuned **all-MiniLM-L6-v2** and **Logistic Regression** on **GoEmotions** for **6-class** emotion classification; achieved **82% accuracy** after targeted augmentation to resolve class imbalance.
- Built **FastAPI** backend with mood detection & recommendation endpoints; containerized via **Docker**, deployed on **Hugging Face Spaces**.
- Integrated **Last.fm API** for mood-to-genre mapping; **React** frontend deployed on **Vercel**.

Experience

MERN Stack Intern

Mar 2026 – Jun 2026

Tamana | tamanaindia.com

React, Node.js, MongoDB, Express

- Reduced page load time from **8s to 2s** via lazy loading, image compression, and removal of unused dependencies.
- Built **View All Products** page and **Best Selling Products** carousel; improved component structure.
- Extended **MongoDB** product schema with additional feature fields and updated existing records.

Achievements & Certifications

- **Co-Head – Documentation Team & Innovate Sphere Hackathon, Computer Engineering Dept, AISSMS COE** – Led a 10-member documentation team, standardizing cross-team communication workflows; managed execution and logistics for a hackathon with 100–150 participants.
- **Certifications:**
 - Fundamentals of LLMs – The LLM Course, Hugging Face
 - Python for Data Science and Machine Learning Bootcamp – Udemy (25 hrs)
 - The Complete Full-Stack Web Development Bootcamp – Udemy (62 hrs)

Education

Bachelor of Engineering in Computer Science (AIML Honors)

2023 – 2027 (Expected)

AISSMS College of Engineering, Pune

CGPA: 9.01 / 10